

sensible suggestion

B1

[3]

4 (a) (i) $GMm \{(R + h_1)^{-1} - (R + h_2)^{-1}\}$
 $\frac{1}{2}m \{v_1^2 - v_2^2\}$

B1

B1

[2]

(b) $2M \times 6.67 \times 10^{-11} \{(26.28 \times 10^6)^{-1} - (29.08 \times 10^6)^{-1}\} = 5370^2 - 5090^2$
 $M \times 4.888 \times 10^{-19} = 2.929 \times 10^6$
 $M = 6.00 \times 10^{24} \text{ kg}$

B1

C1

A1

[3]

(If equation in (a) is dimensionally unsound, then 0/3 marks in (b), if dimensionally sound but incorrect, treat as e.c.f.)